

Field and Lab Methods for Assessing the Function-Based Condition (Tier III) of Eelgrass (*Zostera marina*) Beds and Meadows in Southern California Estuaries and Embayments

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Introduction

The goal of this document is to describe our recommended methods for collecting eelgrass measurements and samples from the field and processing samples to produce data that can be used in assessing the condition of a given bed. This methodology was conceived and field-tested as a part of SCCWRP's ongoing development of a condition assessment framework for submerged aquatic vegetation (SAV) in California that is focused on interpreting SAV condition from the perspective of a bed's ability to support its natural ecological functions (McCune et al. 2020).

The general assessment framework was designed to be applicable to all species of SAV, with a common approach (i.e., transect surveys to collect combinations of structural metrics that can be used to infer functional capacity) applied across species and habitats to maximize comparability of results. However, given the morphological differences in individual plants and beds/meadows due to the species-to-species differences in autecology (e.g., *Zostera marina* vs. *Zostera pacifica*), as well as the influence of the habitat where the SAV is growing (e.g., embayments vs. open coastline), our goal is to produce a suite of field and laboratory methods for distinct species-habitat combinations.

This report is focused on the sampling and analysis of eelgrass (*Zostera marina*) growing in embayments. This particular species-habitat combination was prioritized due to eelgrass' prevalence in the high salinity portions of Southern California's coastal embayments and the emphasis placed on eelgrass with regards to California's eelgrass mitigation policy, its designation as essential fish habitat, and its association with attaining the Estuarine beneficial use designation (United States Environmental Protection Agency 2010; NOAA National Marine Fisheries Service 2014; California Regional Water Quality Control Board San Diego Region 2020). The underlying structural metrics that are recommended for collection, as well as the associated methods of collection, are based upon recommendations from a panel of SAV experts. These metrics were selected through a literature review and consultation with an expert panel to best represent the key ecological functions that healthy eelgrass beds should provide to the coastal ecosystem (McCune et al. 2020). It is important to note that the data that are produced are not meant to be direct measures of ecological function, but rather structural measures of individual plants and their beds that can be used to infer ecosystem functioning.

Though this specific set of methods have been designed to for application in *Z. marina* beds in embayments, we feel it is reasonable to apply them to *Z. marina* in other habitats or to other closely related species (e.g., *Z. pacifica*). We would not recommend applying these methods to other SAV species like *Ruppia* spp. or *Phyllospadix* spp. without considering the morphological and ecological differences between those species and *Zostera* spp.. Attendant with the field and lab methodology, we also provide quality assurance and quality control recommendations where applicable. However, given the relative newness of this type of standardized data collection, we have yet to amass enough data to fully characterize the natural variance in all of the different metrics. Consequently, we do not provide quantitative data quality objectives (DQO) for most of the data types collected. For more established data types (e.g., macrobenthic infauna composition) DQO targets are suggested along with the QA/QC recommendations. As surveys of eelgrass condition using a standardized suite of techniques and data types increase in frequency, we anticipate eventually being able to create quantitatively informed DQO guidelines in future iterations of this methods document.

Overview

This document describes a sequence of events to be done in sampling an individual eelgrass bed. Beginning with reconnoitering the sites, collecting field measurements, collecting samples for laboratory analysis, and conducting laboratory analyses. The instructions have been composed for sampling subtidal beds, though presumably the methods could be adjusted for sampling intertidal eelgrass beds. Furthermore, the primary instructions are constructed to work within the context of a regional-scale, randomly selected sampling scheme. It is assumed that field crews will consist of three people (two underwater and one as surface support). Smaller beds can be sampled by a two-person field crew, as can larger beds if time is available. Conversely, sampling crews greater than three people working together may expedite the sampling of larger beds.

The primary description of each step is framed around the assumption of the sample site being a discrete/semi-discrete eelgrass bed oriented along the shoreline; between 10 - 50m long (parallel to shore) and 5 - 20m wide (shallow to deep). Suggestions for altering certain parts of the protocols to better fit smaller patches of eelgrass or larger meadows of eelgrass are provided. However, in these scenarios some degree of judgement on the part of the field crews will be required. Site-to-site variability highlights the importance of site reconnoitering prior to sampling, which should allow field crews to discuss potential method adjustments with the project leader to ensure all parties on the same page. The field- and lab-based measurements to be made are listed in Table 1. Additional data and samples that could be collected are listed in Table 2.

Table 1 Eelgrass plant and bed-scale metrics to be measured as part of this protocol and which step details their collection.

Metric	Field or Lab Data Generation	SOP Step
Above Ground Biomass	Lab	Step 19
Above Ground TOC & TN	Lab	Step 21
Bed Perimeter	Field	Step 4
Bed Area:Perimeter	Field	Step 4
Below Ground Biomass	Lab	Step 19
Below Ground TOC & TN	Lab	Step 21
Epiphyte Biomass	Lab	Step 16
Leaf Area	Lab	Step 18
Leaves per Shoot	Lab	Step 17
Infauna Diversity	Lab	Step 22-2
Infauna Biomass	Lab	Step 22-3
Percent Cover	Field	Step 7-3
Sediment Grainsize	Lab	Step 20
Sediment TOC & TN	Lab	Step 21
Shoot Density	Field and Lab	Steps 7-4 & 14
Shoot Height	Field and Lab	Steps 7-5 & 15

Table 2. Other environmental measures that could be collected during sampling of the eelgrass beds, but whose methods are not detailed in this SOP. Many measures are not required, but would be useful in different programs depending on their goals and capacity. A diurnal frequency of measure implies multiple measurements taken per day across a spring-neap tidal cycle. This is not meant to be an exhaustive list of additional measures one may collect.

Measurements	Optional / Required	Frequency of Measure
Bed Depth	Required	Multiple Points
Dissolved Oxygen	Optional	Point or Diurnal
Irradiance	Optional	Point or Diurnal
pH	Optional	Point or Diurnal
Salinity	Required	Point or Diurnal
Secchi Depth	Required	Point
Water Column chl a	Optional	Point
Water Column eDNA	Optional	Point
Water Column nutrients	Optional	Point
Water Temperature	Required	Point or Diurnal

Collecting Eelgrass Data

This section details a step-by-step procedure of our recommended methodology used in a complete sampling event for monitoring and assessment of eelgrass beds in southern California estuaries. Steps of the procedure are ordered in sequence of events from initial recon of sampling sites to laboratory analytical procedures. An order of events will help to guide users toward a general timeframe for each event. Collection and analysis of these data will require specific equipment and software, partially listed in Table 3.

Table 3. List of equipment needed to collect the eelgrass data as detailed in this SOP. Note that consumables (plastic bags, jars, etc.) and personal equipment (dive bags, etc.) are both included.

Equipment	Type of Use	Associated SOP Step
0.0625 m ² Quadrat	Field	Step 7
0.25 m ² Quadrat	Field	Step 7
2-cm X 6-cm Sediment Corer	Field	Step 9
10-cm diameter ring	Field	Step 8
10-cm X 15-cm Sediment Corer	Field	Steps 9 & 10
500-µm Sieve	Field/Lab	Steps 10, 22
1-mm Sieve	Lab	Step 13
Area Meter	Lab	Step 18
Compound Microscope	Lab	Step 22-2
Coolers/Ice Chests	Field	Steps 8-10

Dive computer w/ Depth, Temperature, and GPS	Field	Steps 4 & 7
Drying Oven	Lab	Steps 13, 16, 17, 20 & 22
Elemental Analyzer (CHN)	Lab	Step 21
Floats and line	Field	Steps 1, 4 & 6
Meter Stick (Short Transect Tape)	Field	Steps 7 & 15
Microbalance	Lab	Step 12, 13, 16, 19, 20, 21 & 22
Muffle Furnace	Lab	Steps 13, 16, 17 & 22
Sediment Laser Diffractometer	Lab	Step 20
SCUBA Equipment	Field	Steps 4-10
Stereo Microscope	Lab	Step 22-1, 22-2
Waterproof GPS Unit	Field	Steps 1 & 4
Waterproof Metric Transect Tape	Field	Step 6

Step 1 – Site Selection and Reconnoitering

The primary guidance in this step is oriented around a generic regional-scale, random, probabilistic sampling scheme, where discrete sampling locations are randomly selected from a base map or sample frame of existing eelgrass beds/patches (known or suspected presence). These sites are then the basic unit of monitoring and assessment. However, the principles are transferable to other sampling schemes as are illustrated in the modifications section.

Prior to sampling, the geographic coordinates of a site or sites should be identified from the appropriate sample frame using a random or stratified random sampling scheme (e.g., Stevens and Olsen 2004; Olsen et al. 2012). These will typically be provided to the field crews by the project coordinator. Site reconnoitering (recon) can be done with a combination of desktop (aerial imagery, recent mapping surveys, etc) and field-based investigation. Recon should include: 1. Confirmation of the presence of eelgrass at the site; 2. Determination of the approximate size and nature of the eelgrass (patch, bed, meadow, etc.); 3. Classification of the sites for survey purposes; and 4. Identification of access points and potential hazards.

For the purposes of the random, probabilistic sampling design, sites should be classified as **Target Sampled (TS)**, **Target Non-Sampled (TNS)**, or **Non-Target (NT)**. A site would be considered Target Sampled if there is a contiguous eelgrass bed or eelgrass patch within an ~ 200m radius of the initial site coordinates and the site can be sampled. If there is a contiguous eelgrass bed or eelgrass patch within an ~ 200m radius of the initial site coordinates but it cannot be sampled (e.g., access issues, hazardous conditions), it would be classified Target Non-Sampled. If there is no eelgrass within that buffer, or if there is only another type of SAV present (e.g., *Ruppia maritima*), then the site would be classified as Non-Target and would not be considered for sampling. The reason for non-sampling should be recorded and an alternative site (**Overdraw**) should be identified.

Once the presence of eelgrass at the site is confirmed, a rough approximation of the location and extent of the plot of eelgrass to be sampled – the **Sampling Bed** – should be made at the Target sites. Identifying if the sample site is a **Patch** - small, discrete growth of eelgrass less than 3m in diameter – or part of a **Meadow** – very large contiguous growth of eelgrass (e.g., in excess of 75m X 20m) – will be helpful in

planning the time and gear need for the sampling of a given site. This approximation can be done above water or via snorkeling; it is just a rough estimate to inform future field crews. For Target sites, site recon field notes as well as photographs above and below water of the Sample Bed should be collected as and when possible.

During reconnaissance, the accessibility of the sampling site should be evaluated. Logistical considerations should be made for a site – can it best be accessed from shore or best accessed from the water. Furthermore, permit for access should be determined (e.g., private property, protected waterbodies). Any potential underwater or overhead hazards to field crews should be identified (e.g., submerged structures, floating docks, heavy boat traffic, etc). Presence of harmful algal blooms, potential stormwater or septic contamination of water column that may negatively affect field crews should be considered as well. If eelgrass is present, but it cannot be safely accessed or permission cannot be obtained, the site would be considered Target Non-Sampled.

Potential modifications to reconnoitering

For a non-probabilistic sampling scheme, including frequently re-visited sites, site recon is still encouraged, as eelgrass beds can be ephemeral and local conditions may change from year-to-year. Site classifications as Target or Non-Target can be disregarded.

Conversely, a random sampling scheme can be applied in a non-regional context (i.e., within a given embayment or pairs of embayments). In these circumstances the Target, Non-Target classification scheme can still be applied.

Step 2 – Dive Planning and Field Preparation

Nearly all eelgrass beds in Southern California will require use of SCUBA for field sampling. As such, field crews should possess scientific diver training, and each sampling event should include planning for diver emergencies and diver safety. Dive plans are constructed to inform all parties involved in the sampling event of the nearest emergency medical treatment facility, nearest hyperbaric chamber, emergency contacts for harbor patrol and any potential hazards and plans for dealing with hazards. Divers will need to contact harbor patrol and their respective dive safety officer to get approval of dive plans. Prior to departure for sites, an inventory of equipment, as well as weather and water conditions should be checked. In addition, if the location has boat traffic, a surface kayaker is recommended in addition to the two-person dive team. All divers should be in compliance with their institution's diving protocols.

Each member of the underwater field crew should consult percent cover calibrated images prior to sampling to ensure that there is within-crew consistency of judgement-based measures like percent cover. Standardized percent cover images and calibration guides for eelgrass are available from a variety of sources, including Seagrass-Watch (www.seagrasswatch.org/manuals/#guides).

Step 3 – Mapping and Perimeter Measurement

At Target sites, the eelgrass to be sampled – the **Sampling Bed** – must first be mapped. The idealized Sampling Bed should be between 10 and 50m of contiguous /semi-contiguous SAV growth along its longest dimension and 3 to 20m in a perpendicular direction. For larger eelgrass meadows in excess of 50m, a representative 50m X 20m sub-section of the meadow should be identified as the Sampling Bed. If the sampling site (and the 200m buffer) contain distinct patches of eelgrass, an individual patch should be selected for sampling.

Sample Beds are best mapped by two divers and one surface support in a kayak, paddleboard, or other such craft. The perimeter of the sample bed should be measured using a GPS unit. The GPS unit

(contained within a dry bag) is held by the surface support as the divers mark a starting point and follow the edge of the Sample Bed. The surface support should follow the surfacing bubbles or a float towed by the divers, while the GPS unit is actively tracking points into a polygon feature layer (Figure 1.). Points should be tracked at five second intervals automatically by the GPS unit. Once divers reach the starting point marker after swimming the perimeter of the Sample Bed, the GPS should be taken out of tracking mode. Alternatively, if the divers have access to a fully submersible GPS with tracking capabilities, they can eschew a surface-bound GPS. Divers should record water depth at the sediment surface ~ every 5m while swimming the perimeter.



Figure 1. Surface personnel following bubble trails of underwater personnel with GPS tracking unit mounted to a floating board to record the perimeter of the eelgrass bed.

Any notes from divers and surface support regarding deviation from mapped course should be noted and accounted for when processing the map in ArcMaps (ESRI™) or the equivalent. Polygon layers from the GPS unit should be uploaded to a computer with ArcMaps where polygons can be edited. A satellite image overlay should be added as a base map in order to assist divers in referencing landmarks within the bed and for planning transect locations. Within the GIS, **Bed Perimeter** to the nearest 0.5m, and **Bed Area** to nearest 0.25m² should be calculated. **Area:Perimeter** ratio can be subsequently calculated from those measures.

If at any point during the sampling event invasive species are observed at the site (e.g., *Caulerpa* spp.) appropriate California Fish and Wildlife (<https://wildlife.ca.gov/Conservation/Invasives/Species>) and

NOAA offices (<https://www.fisheries.noaa.gov/west-coast/habitat-conservation/caulerpa-species-west-coast>) should be informed. Irrespective of observable invasive species presence or absence, care should be taken to prevent biological contamination from site to site and waterbody to waterbody by cleaning and decontaminating all gear between sampling events (e.g., <https://invasivemusselcollaborative.net/wp-content/uploads/2018/11/NOAA-Decon-Watercraft.pdf>).

Potential modifications to mapping:

In situations where the Sample Bed is a subsample of a larger eelgrass meadow, the normal measurement of the perimeter around that subsample should occur. In addition, a rough mapping of the larger meadow should be done as well. This can be done in the field using the GPS from the surface support craft or from recent remote sensing imagery or other recent mapping efforts.

Mapping QA/QC:

At the start of every field day, the GPS unit should be calibrated following the manufacturer's instructions. Additionally, GPS memory stores should be checked to ensure sufficient memory is available for the beds to be measured on the day.

The perimeter at a minimum of one of every ten beds sampled within a discrete project should be randomly selected apriori to be measured two times. This information will be provided to field crews by the project coordinator. As an example, if four different beds are part of a project, we recommend that one of those beds are randomly selected to be measured twice. Similarly, if there are thirty beds sampled in a given project, we would recommend that three beds be selected for a second perimeter measurement. The second perimeter measurement should serve as a field replicate (and coded as such in data entry) to help quantify method variability.

Step 4 – Sampling Design Planning

Using any spatial data and notes from the site recon in Step 1, approximate transect layout and sampling points should be diagramed to prepare for field sampling. The **Mid Transect** should run the length of the longest dimension of the bed, which generally will be parallel to shore in discrete beds, at a relatively constant depth. The **Cross Transect** should run from the shallowest subtidal portion of the bed through the center of the Mid Transect, unless that is impractical due to abrupt changes in bottom depth. In such cases, the Cross Transect, shallow/deep sampling points should be set up independent of each other. Mid Transect sample points should be established at five equidistant locations from the zero marker (see step 5) to the end point of the primary transect. The Cross Transect sample points should be collected at the shallow and deep ends of the Cross Transect unless, due to the depth variation, a reasonable transect could not be established as noted above.

Ideal Patch

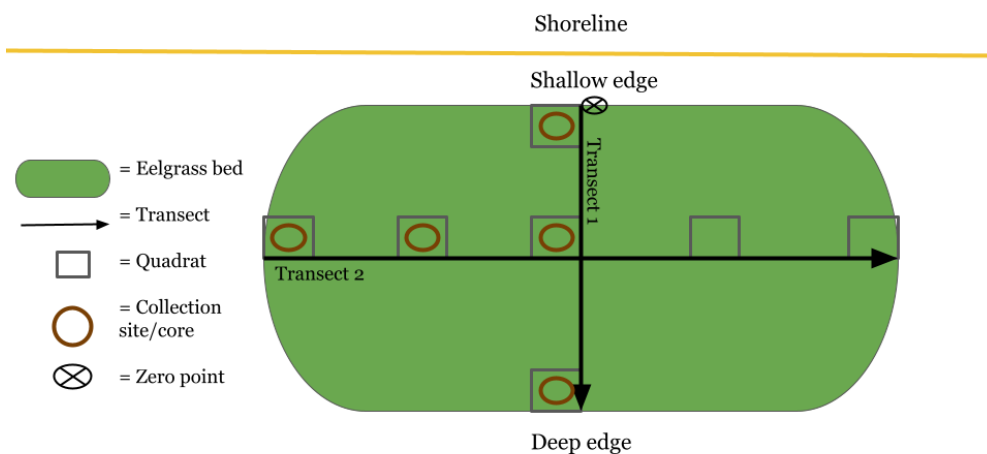


Figure 2 Conceptual diagram of the normal/idealized sampling scheme within an eelgrass bed. Mid Transect and Cross Transect lines are indicated by the black arrows across the middle section of the bed and from the deepest to the shallowest locations within the bed. Quadrat sample locations indicated by grey squares and tissue collection locations indicated by the red circles. Note these are not to scale.

Small Patch

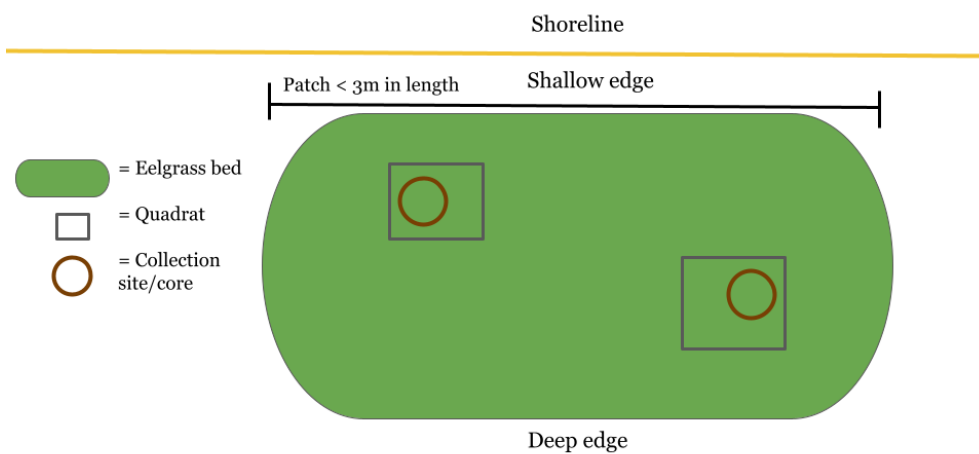


Figure 3 Conceptual diagram of the small patch sampling scheme within an eelgrass bed. Transects are eschewed in favor of one or two haphazardly placed quadrats (grey squares) where all observational measurements are made and from which tissue samples are collected (red circles)

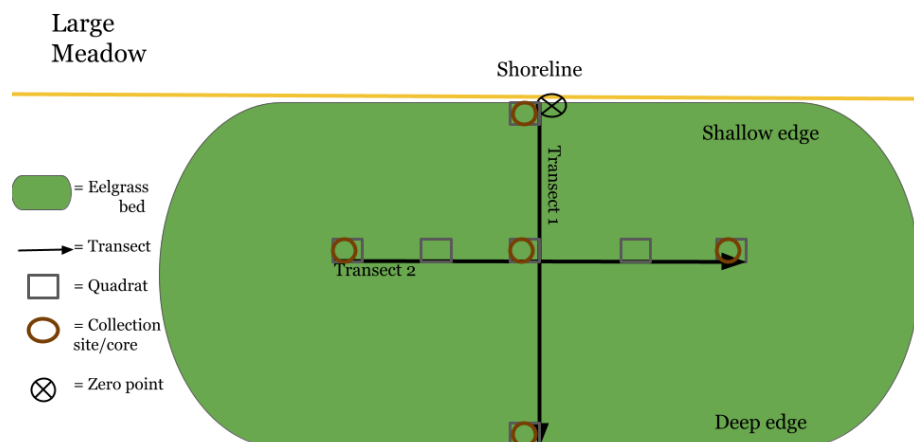


Figure 4 Conceptual diagram of the large meadow, sub-sampling scheme within large contiguous bodies of eelgrass bed. Mid Transect and Cross Transect lines are indicated by the black arrows across the middle section of the bed and from the deepest to the shallowest locations within the subsection of the meadow. Quadrat sample locations indicated by grey squares and tissue collection locations indicated by the red circles.

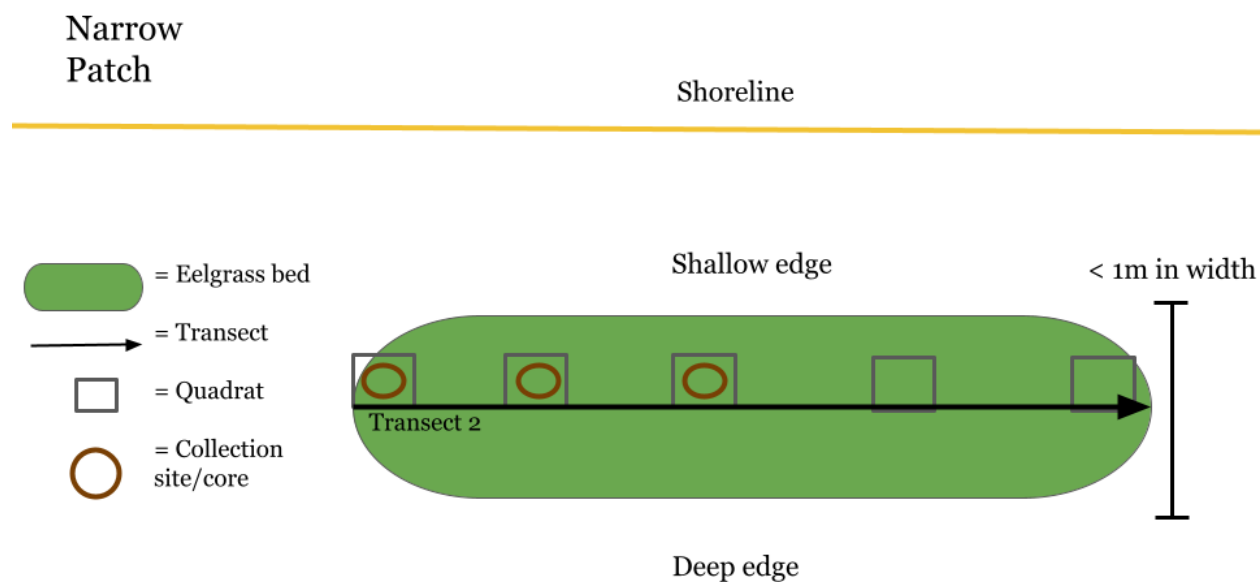


Figure 5 Conceptual diagram of the narrow fringe sampling scheme within an eelgrass bed. The Mid Transect line is indicated by the black arrows across the middle section of the bed. The Cross Transect is omitted. Quadrat sample locations indicated by grey squares and tissue collection locations indicated by the red circles.

Potential modifications to sample design:

1. With small, discrete patches of eelgrass smaller than 3m in diameter, simple placement of 1 or 2 sample points in the middle of the patch can be substituted for the transect-based approach (Figure 2).
2. In large meadows, where the Sample Bed is a subsample of the larger whole, the primary transect should be placed along a consistent depth range (as determined by diver depth gauge during survey) that can be established – Ideally at the median depth of the meadow. The secondary transect should be placed to capture any depth gradient in those beds (Figure 3).
3. In narrow fringing beds where the width is less than 1m and there is little difference in depth across the bed, the Cross Transect can be omitted (Figure 4).

Step 5 – Logger Deployment

If continuous data loggers for environmental conditions (e.g., light, water temperature, salinity, oxygen, current velocity) are to be deployed as part of study, it is recommended that they are done so at least two weeks prior to sampling (e.g., during site recon). These types of activities will be project or study specific. As such, detailed protocols for each such instance should be developed and followed. However, certain generalities for deployment of loggers in concert with eel grass monitoring should be followed. Given the relatively shallow depth of eelgrass beds and their proximity to intensive human activity, using small, relatively inconspicuous loggers or larger loggers firmly anchored to the substrate are recommended. Light and temperature loggers (OnSet HOBOTM Pendant UA-002-08) are likely the most common types of loggers that one might deploy and can be paired at the different depths within SAV beds. Sensors can be attached to buoys that are rigged to weights via line of a length that will allow the buoy/sensor to float above or within the SAV canopy (e.g., Figure 5). Other types of loggers like salinity, pH, or dissolved oxygen could be deployed at a single point within the canopy, as there is a presumption of less meaningful small-scale of variability or attenuation of water quality variables.

In most estuaries and embayments, loggers will need to be cleaned frequently to remove settling sediment and biofilms. It is recommended that at sites with high public accessibility, loggers be swapped with new loggers or data are downloaded during each cleaning to ensure that stolen or lost loggers do not result in significant data losses.

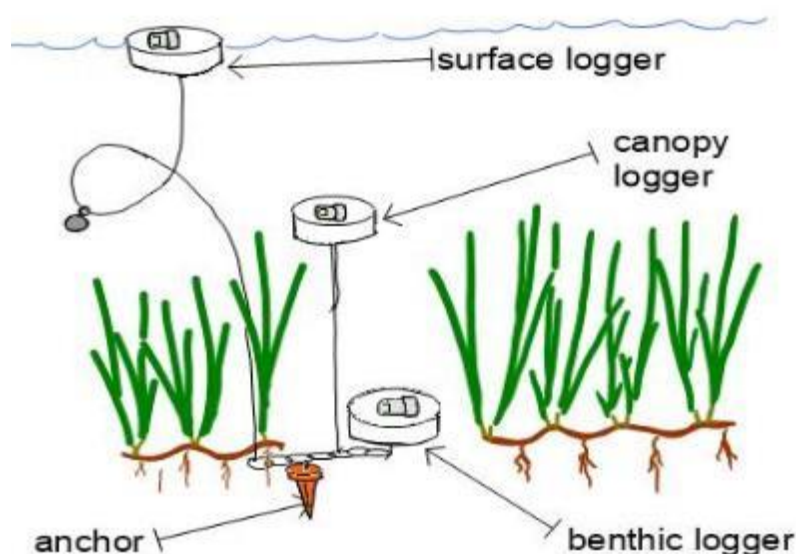


Figure 5 An illustration of an example deployment scheme of paired light and temperature loggers at different points throughout the canopy.

Step 6 – Transect Deployment

Upon entering the water, divers should reacquaint themselves with the location of the eelgrass bed, assess general conditions and water visibility, as well as orient their compasses to the long axis of the Sample Bed. If the long axis is mostly north-to-south – a heading between 315° and 45° or 135° and 225° - then designate the most northern point the zero point of the Mid-Transect. If the long axis is mostly east-to-west – a heading between 314° and 226° or 46° and 134° designate the easternmost point the zero point of the Mid-Transect.

Typically, a transect tape is first laid out across the depth gradient of the bed (conceptually perpendicular to shore and the long axis of the bed) to create the Cross Transect, with the shallow point as the zero point. Record the compass heading of the transect from the zero point. Tent stakes should be used to affix the end of the transect tape on one side of the bed. The Cross Transect should extend up to maximum of ~20m in length.

The Mid-Transect should be deployed along the long axis of the bed across a relatively consistent depth, if possible. Record the heading of the transect from the zero point. During deployment of the transect tape, divers should monitor the depth to ensure that no significant changes in depth occur along the transect – this is also checked when recording observational data. The Mid-Transect should traverse the longest dimension of the bed, as best possible, up to a maximum transect length of ~50m.

Potential modifications to transect deployment

With small, discrete patches of eelgrass smaller than 3m in diameter, transects can be eschewed in favor of haphazard placement of quadrats for sample collection. In large meadows, where the Sample Bed is subsample of the larger whole, the Mid Transect should be placed along any given isobath that can be established – ideally at the median depth of the meadow. The Cross Transect should be placed to capture any depth gradient in those beds. In “narrow” fringing beds where the width is less than 1m there and is little difference in depth across the bed, the Cross Transect can be omitted.

Table 3. Schematic summary of measurements to be made or samples to be collected from each sample collection point within a given eelgrass bed.

Field Measure / Collection of Material	Sample Location						
	mid.1	mid.2	mid.3	mid.4	mid.5	cross.1	cross.2
Percent Cover							
Shoot Density							
Flowering Shoot Density							
Shoot Height							
Water Depth							
Above Ground Material							
Below Ground Material							
Sediment Composition							
Benthic Infauna							

Step 7 – Observational Quadrat Surveys

Two sample points should be established at the shallow (cross.1) and deep (cross.2) ends of the Cross Transect. Five sample points should be established at approximately equidistant points along the Mid Transect, beginning at the zero point of transect (see Step 6) of the transect with sample mid.1 and proceeding to mid.5 (Figure 2). Sample mid.3 should always be the approximate center of the Sample Bed. As an example, if the Mid Transect is 5m long, then a sample would be collected approximately at the 0, 1.25, 2.5, 3.75, and 5m marks. Conversely, in a bed with a 30m transect, samples would be collected approximately at the 0, 7.5, 15, 22.5, and 30m marks. Samples should be a minimum of 1m apart from each other. The types of measurements or samples to be collected in each sample location are summarized in Table 3.

1. At sample locations mid.1, mid.2, mid.3, mid.4, mid.5, cross.1, and cross.2, lay a weighted 0.5x0.5m quadrat down on a predetermined random side (left or right) of the of the transect with the bottom corner of the transect aligned with the determined sample location. The quadrats located at the edges of the bed should be placed as much within the bed as possible.
2. Take a picture of the quadrat from a top-down angle if visibility allows for full view of the transect without losing sight of the sediment surface. Notes should be taken to reflect the intensity of surge or other possible factors that might have interfered with photo representation of the actual quadrat in a way that would affect percent cover estimation – i.e. diver unable to maintain a proper overhead view while taking photos in high surge water.
3. **Percent Cover** is an approximation of the amount of the surface area delineated by the quadrat that is covered by eelgrass vs. bare sediment/macroalgae/other cover. Visual estimates should be made as a continuous value between 0-100% by the diver from a consistent, typically overhead /plan view viewpoint. Before each dive, it is recommended that divers benchmark their estimates to a visual standard (e.g., www.seagrasswatch.org/manuals/#guides)
4. **Shoot Density** is measured by counting the number of shoots within the quadrat. Shoot counts are done by using a string affixed to the quadrat, which is used to fold counted shoots under to keep track of progress through the quadrat (See Figure X). A single shoot is counted by feeling for the stem protruding from the sediment surface. It is not counted as individual leaves, which can branch into multiples at the tip of the stem. If shoot densities are over 100 per 0.25m², a 0.25x0.25m (0.0625m²) quadrat can be used in lieu of the regular quadrat to expedite the process.
5. **Flowering Shoots Density** should be recorded during measurement of total shoot density. If a bed is blooming at time of sampling, the number of shoots with flowers or seed packets/spathes should be noted during the shoot density measurement.
6. **Shoot Height** is measured using a small (<2m long) transect tape or metric measuring stick. One diver holds and moves the measuring device to the base of each of five randomly selected shoots within the quadrat. The other diver measures and records the height. Shoots are measured from the base of the sediment to the tip of the tallest leaf on one shoot per replicate. If the selected leaf contains is flowering, make a note of it. Do not measure multiple leaves on one shoot. If there are less than 5 shoots within the quadrat, there will be less than 5 replicates.
7. Water depth at the sediment surface should be recorded from dive computers for each sample location.

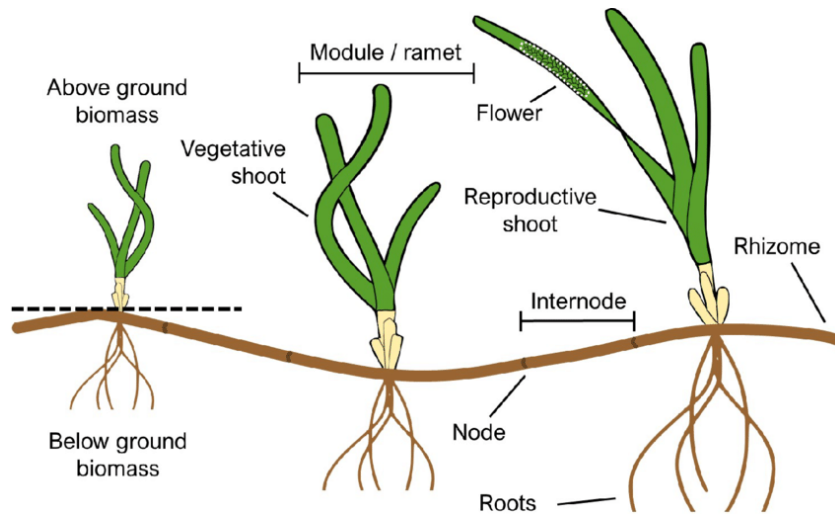


Figure X. Diagram displaying where to measure shoot height. (Howarth, L.M., Lewis-McCrea, L., LaBelle, J., and Reid, G.K., (2021) *Managing Aquaculture and Eelgrass Interactions in Nova Scotia. Centre for Marine Applied Research (CMAR), Dartmouth, Nova Scotia, Canada.*)

Potential modification to observational quadrats:

With small, discrete patches of eelgrass smaller than 3m in diameter, place 1 or 2 quadrats within the eelgrass. If the patch is large enough to place 2 quadrats while maintaining approximately 1m distance between them, do so labelling them samples mid.1 and mid.3. Otherwise use 1 quadrat in the center, as sample mid.3. Collect all measures as noted above.

Observational survey QA/QC

Measures of percent cover, shoot density, flowering shoot density, and shoot height should be measured independently by both divers to capture measurement variance. Data should be designated as Diver A or Diver B during data entry.

At a minimum, 10% of the total number of beds sampled during a project should be resampled as a field replicate with the same or different pairs of divers on the same day of sampling. As an illustration, if 7 beds are sampled during the course of a project, we recommend that 1 bed should apriori be randomly selected to be resampled. Similarly, if 25 beds are sampled during the course of a project, 3 should be randomly selected for resampling.

If the randomly selected bed is wide enough, the Mid and Cross transects can be re-laid and all of observational measures, including Diver A and Diver B measures, can subsequently be collected to serve as a field replicate (and coded as such in data entry) to help quantify method variability. If the bed is too narrow to place another set of transects through a different part of the bed, the same transect can be used for the field replicate. However, in this instance the quadrats should be placed on the opposite side of the transect as they were for the original measure. As an illustration, if the quadrat at sample mid.1 was placed to the right of the transect line during the initial sampling, then the replicate would be placed to the left.

Step 8 – Above Ground Collections

At sample locations mid.1, mid.3, mid.5, cross.1, and cross.2, above ground eelgrass tissue should be collected and placed in pre-labeled, sealable plastic bags. Ideally sample bags will have both an interior and exterior label. The goal is to collect all material from a known area (e.g., g cm⁻²)

1. Place a 10-cm (i.d.) ring on the sediment surface within the quadrat to demarcate the area to be sampled. All shoots within the ring are to be collected. The sample location should contain at least one shoot and there should be little to no obvious obstacles for the below ground biomass core to penetrate the sediment – e.g., large rocks or shelled invertebrates – that could prevent subsequent collection of below ground material.
2. One diver holds the plastic bag, while the other diver separates all above ground shoots within the ring at the upper most rhizome nodule at or near the sediment surface. Above ground material can be clipped using a cutting device or broken off by hand.
3. Shoots are gently placed into labeled sample bags. With long shoots, gently rolling the SAV tissue into a coil will help to make the tissue fit. Before sealing, most of the water within the bag should be pushed out carefully without pushing out SAV tissue. This will help reduce the sample weight when divers exit the water, as a potential total weight of all sample bags filled with water can exceed 40 lbs.
4. Divers can elect to place the plastic sample bags in mesh dive bag and retain all of the samples during sampling. Alternatively, samples can be given to surface support crew, or they can be marked with a pop-up surface buoy and collected after sampling.
5. Once samples are removed from the water, the seals should be checked on the bags, the internal and external labels should be confirmed that they are intact, and then the bags should be placed into a cooler.

Potential modifications to above ground collection

At small, discrete patches less than 3m in diameter, above ground material should be collected at all (i.e., 1 or 2, depending on size) of the quadrats placed in the patch. In narrow beds less than 1m wide, the cross.1 and cross.2 samples should be omitted, as noted above.

Above ground QA/QC

At a minimum, 10% of the total number of beds sampled during a project should be resampled as a field replicate. As an illustration, if 7 beds are sampled during the course of a project, we recommend that 1 bed should apriori be randomly selected to be resampled. Similarly, if 25 beds are sampled during the course of a project, 3 should be randomly selected for resampling.

If the randomly selected bed is wide enough, the primary and secondary transects can be re-laid and all the above ground samples can subsequently be collected to serve as a field replicate (and coded as such in data entry) to help quantify method variability. If the bed is too narrow to place another set of transects through a different part of the bed, the same transect can be used for the field replicate. However, in this instance the material should be collected from the opposite side of the transect as they were for the original measure. As an illustration, if the quadrat and subsequent tissue collection at sample mid.1 were placed to the right of the transect line during the initial sampling, then the replicate would be placed to the left.

Step 9 – Below Ground Collections

After above ground tissue is collected, the associated below ground tissue should be collected from the same above ground biomass footprint in sample locations mid.1, mid.3, mid.5, cross.1, and cross.2. The goal is to collect all material from a known area (e.g., g cm^{-2}). Additionally, a small core for sediment grainsize, TOC, and TN content should be collected within the quadrat.

1. Using the same 10-cm (i.d.) ring, all below ground eelgrass material (i.e., roots and rhizomes) within the footprint of the ring must be removed, typically to a depth of 10+cm. A variety of approaches can be used to remove the material, but care should be taken to minimize the inclusion of rhizomes that run outside the footprint of the ring.
 - a. A metal trowel can be used to cut the rhizomes within the core and excavate the eelgrass and associated sediment
 - b. A pair of scissors or other cutting device can be used to clip the rhizomes along the perimeter of the ring and rhizomes pulled out
 - c.
2. Below ground material and any associated sediments should be placed into a plastic bag with interior and exterior labels.
3. A small core ~3cm i.d., should be inserted approximately 5cm into another, preferably unvegetated portion of the quadrat for **Sediment Analysis**. A 50-cc syringe with the bottom cut off works well for this. *Note that if a hydrometer is to be used for sediment analysis, multiple small cores of sediment will be needed. The project coordinator should be consulted prior to the beginning of the project.*
4. The Sediment Analysis core should be pulled from the sediment and the contents extruded into a small sealable plastic bag with interior and exterior labels.
5. Once the samples are placed in their respective plastic bags, they are stored in a mesh dive bag for temporary transport throughout the rest of sampling or transferred to surface support crew and placed in a cooler.

Potential modifications to below ground collection

At small, discrete patches less than 3m in diameter, below ground material should be collected at all (i.e., 1 or 2, depending on size) of the sample sites in the patch. In narrow beds less than 1m wide, the cross.1 and cross.2 samples should be omitted, as noted above.

Below ground QA/QC

At a minimum, 10% of the total number of beds sampled during a project should be resampled as a field replicate. As an illustration, if 7 beds are sampled during the course of a project, we recommend that 1 bed should apriori be randomly selected to be resampled. Similarly, if 25 beds are sampled during the course of a project, 3 should be randomly selected for resampling. The same sample locations used for the above ground field resampling should be used for the below ground resampling.

Step 10 – Infauna Collection

A 10cm(i.d.) X 15cm long core should be used to collect a sediment sample for infauna. The sample should be collected near the center of the bed, though it does not have to be within the quadrat associated with sample mid.3.

1. Identify a location near sample quadrat mid.3 and insert the core a minimum of 10cm into the sediment with a twisting motion to cut through the rhizomes. Remove as much of the above ground vegetation as possible without pulling up surface sediment.

2. The core should be wiggled in a circular manner to dislodge surrounding sediment and allow enough space for the diver to reach a hand down to the bottom of the core and cover the bottom and prevent the contents from dislodging.
3. Removal of the core should be done slowly while the bottom of the core is covered, then divers may move slightly adjacent to the core location before pushing the contents of the core into a plastic bag with interior and exterior labels held by the other diver. Moving slightly away for the transect might allow for enhanced visibility, as the sediment coring process can disturb sediment and reduce visibility in the immediate area.
4. Upon completion of all of the diving tasks, the material in the bag may be gently sieved on a 500µm screen in ambient water (see alternatives below). Make sure to get all of the material from inside the plastic bag. When sieving, being quick will help to prevent animals from wrapping themselves within the screen and being gentle will minimize damage to the fauna from the sloshing of sand and shells. Sieved material should then be placed in a screw top plastic jar, with care taken to make sure animals and detritus are not retained on the screen. Squeeze bottles and forceps can be helpful to ensure collection of all animals. Jars should be placed on ice until they can be chemically preserved in the laboratory. Alternatively sieved material can be placed back into the plastic bag to be transferred to a jar in the laboratory.
 - a. We recommend sieving material in the field, but it can be done upon returning to the laboratory. If material is to be sieved in the laboratory, the sample bags must be kept on ice to minimize decomposition of the fauna in the core.
 - b. Whole core preservation....

Potential modifications to infauna collection

At small, discrete patches less than 3m in diameter, the core for infauna can be taken in the approximate center of the patch.

Infauna collection QA/QC

At a minimum, 10% of the total number of beds sampled during a project should be resampled as a field replicate. As an illustration, if 7 beds are sampled during the course of a project, we recommend that 1 bed should apriori be randomly selected to be resampled. Similarly, if 25 beds are sampled during the course of a project, 3 should be randomly selected for resampling. The resample core for infauna should be collected near the center of the bed, but away from sediments disturbed during other sampling activities. The resample core(s) should be used quantify method variability.

Step 11 – Field Debriefing

One diver moves samples to a cooler with ice immediately after divers exit the water and remove gear, while the other diver records sample conditions and checks data recordings for clarity:

1. Divers exit the water and remove SCUBA gear while ensuring that sample bags are carefully placed aside and not punctured. Inspect for leaking and have both field personnel work on fixing leaks by double bagging samples quickly.
2. Samples are placed over ice inside a large cooler. A cooler with at least 15-gallon capacity is recommended to fit all sample bags filled fully.
3. Field conditions – water temperature, visibility, air temperature, sea surface state, weather conditions – should be recorded.

4. All underwater personnel should review and discuss the data recorded on the dive slates to ensure that underwater handwriting is legible. Data can be transferred to waterproof data sheets, if time and conditions allow
5. A photo should be taken of the dive slate(s) with all data on it visible, including date and site information. The photo should be emailed or uploaded to a cloud storage drive for secure storage of the data and ease of access for data entry personnel.
6. The Field Collection Inventory Sheet should be filled out.

Step 12 – Sample Sorting and Preparation

To prevent degradation and breakdown of the eelgrass material, samples should be taken back to the laboratory as soon as possible to be sorted, cleaned and stored.

Following fieldwork

1. Sediment for grainsize, TOC, and TN analysis can be refrigerated for up to 48 hours before processing or frozen (-20°C) for long-term storage.
2. Jars containing sieved benthic infauna (with detritus) samples should be fixed with 20% buffered formalin. Add a volume of formalin at approximately a 2:1 to 3:1 volumetric ratio of formalin to sample. If there are large amounts of organic matter in a sample (e.g., roots, rhizomes, and detritus) try to use a 3:1 ratio. After 5-7 days of exposure to formalin, transfer the sample to 70% ethanol, decanting over a 500µm sieve. Store the sample in a dark, room temperature location.
 - a. If infauna are to be identified using genetic techniques, samples should be preserved with a mixture of 95%, non-denatured ethanol and glycerin, mixed at a volumetric ratio of 95:5 instead of formalin. Add the ethanol mixture to the sieved material at a 3:1 volumetric ratio. After 24hrs, replace the ethanol mixture with fresh 95% ethanol (no glycerin), decanting through a 500µm sieve. Replace the ethanol again after an additional 24hrs. Store the sample in a dark, room temperature location.
3. Aboveground and belowground material should be placed in the refrigerator for no longer than 24 hours.
4. All sample labels should be double checked for proper labeling and confirmed with the dive team before departing for the day.

Before sample processing

1. Above ground tissue samples should be checked for appropriate sample content – the sample should not have any belowground tissue (i.e., roots and rhizomes). Any below ground material present in the bag should be sorted out and placed in the appropriate below ground sample. Above ground samples should be placed back on ice or into a refrigerator for no longer than 24hrs.
2. Labeled weighboats for each sample location should be prepared for below ground tissue, above ground tissue, and epiphytes.
 - a. Weighboats for below and above ground biomass can be sheets of heavy-duty aluminum foil or large commercial aluminum weighboats. Weighboats for epiphyte biomass can be smaller. All weighboats should be combusted at 550 °C for 6hrs. Label each boat with a pencil prior to combustion, indenting the aluminum such that the label can be read after the pigment is combusted.
 - b. After cooling, the sheets or boats should be weighed on an electric balance to the nearest 0.1mg and the value recorded as the pan weight.

- c. Note that samples of the above ground and below ground tissue may be bigger than typical commercial weigh boats. In these instances, homemade weighboats can be created with heavy duty aluminum foil crimped to form a container.

Step 13 – Below Ground Biomass

Biomass of the below ground tissue should be measured as wet mass, dry mass, and ash free dry mass to the nearest 0.1 mg using an electronic balance. In between mass measurements, samples should be kept in a lidded container to prevent contamination. If any measurement value is less than the detection limit of the balance, record the smaller of 0.1mg and the detection limit of the balance as the mass value. If measuring frozen samples, allow them to completely thaw in covered conditions.

1. Below ground tissue should be removed from their accompanying sediment by carefully sifting through sediment with hands and pulling out the rhizomes and attached roots. Gently rinse off any attached sediment from the rhizomes over a 1-mm screen with tap water. Place the tissue onto the appropriate weighboat. Rinse all remaining sediment in the sample bag through the 1-mm sieve and pull out any visible root or rhizome tissue. Add them to the labeled weighboat.
 - a. Only intact, live rhizome/root structures should be measured. Many dead or decaying rhizomes may be present in the form of soft, dark material that can be removed from the sample prior to measuring the mass. Live rhizomes will have a slightly orangish-brown appearance when fresh and have a degree of rigidity. However, if there is doubt as to the live/dead nature of a rhizome, keep it in the sample; do not remove it.
 - b. Pat all tissue dry with a paper towel
 - c. Below ground samples can be measured fresh, if measurements are made within 24hrs of collection – being kept in a refrigerator in the interim – or they can be frozen (-20°C) after cleaning off the associated sediment. If freezing, sort out any dead or decaying rhizomes before placing the material in a sealed plastic bag after cleaning instead of the weighboat.
2. Cleaned below ground tissue should be patted dry with paper towels, placed in the appropriate combusted and labeled weighboat for that sample. Measure the contents of the weighboat. Subtract the pan weight from this value to obtain the **Below Ground Wet Mass**.
3. Once wet mass has been measured, place the tissue+weighboat into a drying oven at 60°C for a minimum of 48hrs. After the minimum 48hrs, remove samples from the drying oven and allow to cool in a desiccator. Measure the contents of each weighboat. Subtract the pan weight from this value to obtain the **Below Ground Dry Mass**.
 - a. If eelgrass TOC and TN content are to be measured, remove 2.5-3mg of dried tissue prior to combusting the sample for calculation of ash free dry mass. Record the amount reserved for TOC/TN analysis.
4. Once dry mass has been measured, place the tissue+weighboat into a muffle furnace at 550°C for 6hrs. Allow samples to cool in the oven and then move to a desiccator. Measure the contents of each weighboat. Subtract the pan weight from this value to obtain the ash mass. Subtract the ash mass from the dry mass to obtain the **Below Ground Ash Free Dry Mass**.
5. Once calculation of ash free dry mass has been made for the sample, calculate the ratio of ash free dry mass to dry mass. Multiply the mass of tissue reserved for TOC/TN analysis by the ash:dry ratio and add that value to the ash free dry mass value calculated in Step 13.3 to produce a **Corrected Below Ground Ash Free Dry Mass** value corrected for the removal of tissue between drying and ashing. The use of the “corrected ash free dry mass” value is important to accurately scale the sample measure up to areal or density-based estimates or comparison with samples for which TOC/TN content was not assessed.

Below ground biomass QA/QC

Calibrate the balance following manufacturer's instructions prior to each day of use.

For Steps 14 -18, we recommend working with one sample bag at a time. Labeling and organizing samples for Step 19 – Aboveground Biomass.

Step 14 – Epiphyte Biomass

Epiphytic material should be carefully removed from all the leaves in each sample. Shoots and leaves should remain intact to be measured in Steps 15-18. Epiphyte biomass should be measured as wet mass, dry mass, and ash free dry mass to the nearest 0.1 mg using an electronic balance. In between mass measurements, samples should be kept in a lidded container to prevent contamination. If any measurement value is less than the detection limit of the balance, record the smaller of 0.1mg and the detection limit of the balance as the mass value.

Epiphytic material should be removed from both sides of each leaf using the edge of glass microscope slide.

1. Gently shake the aboveground biomass content in the sample bag and pour the sample bag contents into a 63µm sieve. Then rinse the bag with water and pour the remaining water from the bag into the sieve
2. Using a squirt bottle, squirt water onto the sieve to collect the solid material into one corner of the sieve. Remove all the material off of the sieve into a large petri dish.
 - a. Thoroughly wash the sieve in between samples.
3. Slowly and firmly run the edge of the glass slide from base of each leaf to the tip, making sure to be especially gentle around areas with perforations or tissue degradation on the leaves.
4. Record any clear instances of wasting disease. – see http://www.seagrassli.org/ecology/wasting_disease.html
5. Contents of the scraping should be placed in a small petri dish, examined for general contents and sorted to remove any pieces of eelgrass.
6. Epiphytic matter should then be placed in the appropriate combusted and labeled weighboat for that sample. Measure the contents of the weighboat. Subtract the pan weight from this value to obtain the **Epiphyte Wet Mass**.
7. Once wet mass has been measured, place the material+weighboat into a drying oven at 60°C for a minimum of 48hrs. After the minimum 48hrs, remove samples from the drying oven and allow to cool in a desiccator. Measure the contents of each weighboat. Subtract the pan weight from this value to obtain the **Epiphyte Dry Mass**.
8. Once dry mass has been measured, place the material+weighboat into a muffle furnace at 550°C for 4hrs. Allow samples to cool in the oven and then move to a desiccator. Measure the contents of each weighboat. Subtract the pan weight from this value to obtain the ash mass. Subtract the ash mass from the dry mass to obtain the **Epiphyte Ash Free Dry Mass**.

Epiphyte QA/QC

Calibrate the balance following manufacturer's instructions prior to each day of use. With every 10% of samples, a second person should inspect the leaves of a sample to make sure that all epiphytic material was removed and that leaf tissue was left relatively intact.

Step 15 – Shoot Count

Shoots from the sorted above ground tissue samples should be removed from their sample bag and gently laid out onto a flat surface. The total number of intact shoots in an above ground tissue sample should be

recorded (Figure x). Each shoot within a sample should then be assigned a Shoot ID using a *Sample Number – Shoot Number* format to keep track of it across all shoot-based measurements – e.g., mid.1-Shoot 1, mid.1-Shoot 2, etc. All loose leaves in the sample should be placed to the side until it is time to measure above ground biomass.

Step 16 – Leaves per Shoot

The number of individual leaves should be counted for each shoot (Figure XX). This value should be recorded as the measure of **Leaves per Shoot**. Measurements should be tracked using the shoot ID scheme established in Step 15.

Step 17 – Shoot Height and Width

The length of each intact shoot from its base to the tip of its longest intact leaf should be measured to the nearest 0.1mm (Figure XX). The width of the shoot should be measured in the middle of the shoot. When recording the measurements, the Shoot ID scheme established in Step 15 should be used to track the data.



Figure XX. An example of an eelgrass shoot with 4 intact leaves and a leaf length of 124.1 mm.



Figure XX. An example of an eelgrass leaf and a width of 2.0mm.

Step 18 – Leaf Area

After Steps 14-17, the leaves from each shoot in a sample should be removed and used to measure leaf area to the nearest 1.0mm². If the leaves have dried, then they should be rewetted. Leaf area can be calculated using a leaf area meter or a calibrated bench-top scanner and area estimating software (e.g., <https://imagej.nih.gov/ij/download.html>). Measurements should be tracked using the Shoot ID scheme established in Step 15.

1. Cut the longest leaf within the shoot at the basal end. Cuts should be made at the top of the blade sheath, which is located above the stem and below the base of each leaf. Set the remaining leaves aside for subsequent measure of above ground biomass.
2. Perforated or broken leaves should not be measured.
3. Separated leaves can be placed in a small amount of seawater in a large petri dish to keep the leaves from wilting.
4. Wetted leaves should be blotted dry before measuring area.
 - a. Area can be measured with a leaf area meter (LI-COR™ LI-3100C Area Meter). Make sure each blade is laid flat longwise when running through the meter sensor. Three runs are done for each leaf and the mean is reported as **Leaf Area** to the nearest 1.0mm².
 - b. Area can also be measured by scanning the leaves on a benchtop scanner and a 10mmX10mm scale block. **Leaf Area** can then be calculated from the calibrated images using a standard image analysis software package.
5. After measuring the leaf area, place the measured leaf with the rest of the leaves for subsequent measuring of above ground biomass.
6. After epiphyte removal and measurement of leaf area, above ground tissue can be frozen (-20°) if biomass measurements are not going to be completed within 24hrs of collection.

Step 19 – Above Ground Biomass

Biomass of the above ground tissue should be measured as wet mass, dry mass, and ash free dry mass to the nearest 0.1 mg using an electronic balance. In between mass measurements, samples should be kept in a lidded container to prevent contamination. If any measurement value is less than the detection limit of the balance, record the smaller of 0.1mg and the detection limit of the balance as the mass value. If measuring frozen samples, allow them to completely thaw in covered conditions.

1. Above ground tissue with the epiphytes removed should be patted dry with paper towels (if leaves have dried, then leaves should be rewetted), placed in the appropriate combusted and labeled weighboat for that sample. Measure the contents of the weighboat. Subtract the pan weight from this value to obtain the **Above Ground Wet Mass**.
2. Once wet mass has been measured, place the tissue+weighboat into a drying oven at 60°C for a minimum of 48hrs. After the minimum 48hrs, remove samples from the drying oven and allow to cool in a desiccator. Measure the contents of each weighboat. Subtract the pan weight from this value to obtain the **Above Ground Dry Mass**.
 - a. If eelgrass TOC and TN content are to be measured, remove 2.5-3mg of dried tissue prior to combusting the sample for calculation of ash free dry mass. Record the amount reserved for TOC/TN analysis.
3. Once dry mass has been measured, place the tissue+weighboat into a muffle furnace at 550°C for 6hrs. Allow samples to cool in the oven and then move to a desiccator. Measure the contents of each weighboat. Subtract the pan weight from this value to obtain the ash mass. Subtract the ash mass from the dry mass to obtain the **Above Ground Ash Free Dry Mass**.
4. Once calculation of ash free dry mass has been made for the sample, calculate the ratio of dry mass to ash mass. Divide the mass of tissue reserved for TOC/TN analysis by the dry:ash ratio and add that value to the ash free dry mass value calculated in Step 19.3 to produce a **Corrected Above Ground Ash Free Dry Mass** value corrected for the removal of tissue between drying and ashing. The use of the “corrected ash free dry mass” value is important to accurately scale the sample measure up to areal or density-based estimates or comparison with samples for which TOC/TN content was not assessed.

Step 20 – Sediment Grainsize

Grainsize should be measured using the laser refractometer method (e.g., Beuselinck et al. 1998), which can be done either “in house” or via a contracting laboratory. If a laser refractometer is not available, the pipette (e.g., Plumb 1981) or hydrometer (cite) method is acceptable. When using a contracting laboratory, make sure to discuss the format (frozen, wet, dry, etc) and volume of sediment needed by the lab for grainsize analysis. When recording the data, the method must also be recorded.

If sediment TOC and TN are to be measured, a portion of sediment should set aside prior to grainsize analysis. This sediment should be dried in a drying oven at 60°C for at least 48hrs. The goal is to produce between 2.5 and 3g of dried sediment for TOC/TN analysis. Depending on the bulk density of the fresh sediment, the amount needed to set aside will vary.

Sediment grainsize QA/QC

A minimum of one grainsize sample per analytical batch should be selected for reanalysis. These lab replicates are used to quantify method variability. They should be indicated as lab replicates when entering the data. All equipment (e.g., refractometer or electronic balance) should be calibrated according to manufacturers instructions with each batch of samples analyzed.

1. From grainsize data, **%Sand**, **%Silt**, and **%Clay** should be calculated from the phi sizes – if using the laser method – or weighed from the appropriate dried aliquots – if using the pipette method.
2. For each of the lab replicate samples, the relative percent differences (RPD) for each of sand, silt, and clay values between the two replicates should be calculated by:

$$RPD = \{[Absolute(\%Sand_{Rep1} - \%Sand_{Rep2})]/[(\%Sand_{Rep1} + \%Sand_{Rep2})/2]\} * 100$$

3. If the RPD is greater than 25% for any of the three measures, then those measures within the entire batch should be flagged as highly variable

Step 21 – Total Organic Carbon and Total Nitrogen

Total organic carbon (TOC) and total nitrogen (TN) content of the bulk sediment and eelgrass tissue should be measured via combustion with an elemental analyzer. This can be done either “in house” or via a contracting laboratory. Typically ~2.5g of acidified and dried material are needed for analysis, though contract lab’s specifications and manufacturer’s protocols may differ. Make sure to adjust the sample portioning of dried eelgrass (Steps 13 and 19) and sediment (Step 20) based upon analytical requirements. **Sediment TOC and TN, Above Ground TOC and TN, and Below Ground TOC and TN** should be reported in the nearest 0.01mg g⁻¹. Sample type should be indicated as sample and matrix as sediment or SAV tissue, as appropriate.

TOC and TN QA/QC

A minimum of one TOC/TN sample for sediment per analytical batch, as well as for above ground and below ground eelgrass tissue, should be selected for reanalysis. These lab replicates are used to quantify method variability. They should be indicated as lab replicates when entering the data. The elemental analyzer should be calibrated following the manufacturer’s recommendations. Once per 20 samples or once per laboratory batch, whichever is more frequent, standard reference material for TOC and TN should be analyzed to provide a check of data accuracy. Sample type should be indicated as standard.

1. The relative percent difference (RPD) for each of TOC and TN between laboratory replicates should be calculated by

$$RPD = \{[Absolute(TOC_{Rep1} - TOC_{Rep2})]/[(TOC_{Rep1} + TOC_{Rep2})/2]\} * 100$$

2. If the RPD is greater than 25% for either of the two measures, then those measures within the entire batch should be flagged as highly variable
3. Percent recovery for each of TOC and TN for each laboratory standard should be calculated by:

$$\% Recovery = (TOC_{measured} / TOC_{expected}) * 100$$
4. If percent recovery is <80 or >120, the analysis should be stopped and an additional reference material sample should be measured.
5. If percent recovery is still outside of the target, then the machine should be recalibrated until suitable performance can be achieved. The 20 samples associated with the reference material should be flagged as inaccurate.

Step 22 – Infauna Analysis

After benthic infauna samples have been transferred to ethanol as described in Step 12, all macrobenthic organisms alive at the time of sample collection should be sorted from the associated detritus, identified, counted, measured for biomass. Taxonomic identification of benthic infauna requires a reasonable amount of specialized skill and equipment. As such, samples can be sorted and identified “in house” or by an external laboratory. Most taxonomic labs are not equipped to calculate infauna biomass.

1. Infauna should be sorted from the detritus, typically under stereo microscope. A sorting sheet like **Appendix 1** should be used to keep track of sorting progress and QA/AC.

- a. All sorting must be done in 70% ethanol, except for sorters where health and safety issues exist, with care taken to ensure that the sample being sorted is always fully covered with alcohol.
- b. If necessary, sorting may be performed using water, but care must be taken to minimize the time when specimens are not in 70% ethanol. Samples may not be left over night in water, as specimens may degrade. Samples must be placed back into 70% ethanol at the end of each day. For large samples, only placing into water that portion to be sorted in a day's time is advisable.
- c. The organisms removed from the sample are sorted into taxonomic lots for subsequent taxonomic analysis. Typically, organisms are sorted into annelids, arthropods, echinoderms, molluscs, and miscellaneous phyla (e.g., cnidarians, nemerteans). However, each laboratory will determine the taxonomic level of sorting adequate to their needs for subsequent sample analysis by their taxonomists.
- d. Remove all individual organisms and fragments from the sample with the exception of nematodes, foraminiferans and planktonic species, or planktonic life stages of benthic organisms. All fragments, such as decapod chelae and legs, should be placed in their respective taxa lots. Sorters are to be instructed "If in doubt, pick it out."
- e. Sorters should not remove animals from their tubes.
- f. Aggregate the taxa lots into one or more sample containers. Each taxa lot should be internally labeled with the station name, sampling date, and sorter's initials. Place an internal label in each vial/container with this information and split number (i.e., 1 of 2, 2 of 2) if more than one container is used. Labels are to be written in pencil on 100% rag-paper.

2. All of the sorted organisms contained within each sample should be identified to the lowest possible taxonomic category and counted.

- a. The goal of the analysis is to provide species level identifications whenever possible. However, because of difficulties in the taxonomy and the lack of expertise within most laboratories the following exceptions are made
 - i. Kinorhynchs are identified to Phylum Kinorhyncha
 - ii. Oligochaete annelids are identified to Subclass Oligochaeta
 - iii. Hirudinean annelids are identified to Subclass Hirudinea
 - iv. Podocopid ostracods are identified to Order Podocopida
 - v. Harpacticoid copepods are identified to Order Harpacticoida
 - vi. Insecta arthropods may be identified to Subclass or Order
- b. The number of organisms reported must account for all organisms in a sample alive at the time of collection. Fragments of bilaterally symmetrical organisms will be identified and counted only if the fragment includes the anterior end of the organism. For radially symmetrical organisms (e.g., ophiuroids, anthozoans) only fragments bearing the majority of the oral disk will be identified and counted. Also, care must be taken to avoid reporting empty mollusc shells or crustacean molts in the data.
- c. Attached parasites and other epibionts may be noted on the bench sheet as present but are not to be reported in the submitted survey data. Ectoparasites of fish such as cymothid isopods, which may be temporary members of the benthic community, are counted and reported in the submitted survey data.
- d. Nomenclature and orthography follows that used in the most recent SCAMIT species list (www.scamit.org).
- e. If an organism cannot be identified to at least the species level and: 1. is identified at

a coarser taxonomic level; 2 there are other organisms in the sample from the same taxon, but identified to a finer taxonomic level; and 3. if the taxonomist cannot determine if the organism is distinct from other sister taxa then the organism should be indicated as excluded from diversity measurements on the taxonomist's bench sheet and "exclude" should be placed in the Exclude data field when submitting final data.

- i. Example 1: Both *Dipolydora* sp and *Dipolydora socialis* are reported in a sample and the taxonomist cannot determine if the specimen reported as *Dipolydora*. sp is distinct from *D. socialis*. Exclude
 - ii. Example 2: An unidentifiable onuphid polychaete is reported as Onuphidae. It is the only member of its family present in the sample. Do Not Exclude
 - iii. Example 3: Both *Modiolus* sp and *Modiolus capax* are reported in a sample. However, the taxonomist is confident that the specimen identified at the genus-level is not *M. capax*. Do Not Exclude
 - f. Identified organisms should be vialled separately from each other with an internal label indicating the taxonomic ID, station name, and taxonomists initials. Labels should be written in pencil on 100% rag-paper, for permanent wet labels.
 - g. Fragments of organisms should be vialled as with the appropriate identified and counted taxa, as best the taxonomist can determine. Fragments that cannot be associated with any identified organisms should be vialled and labelled as misc. tissue.
3. Once all samples have been identified and passed sorting and identification QA/QC, biomass can be measured. Taxon-specific biomass should be measured as wet mass, dry mass, and ash free dry mass to the nearest 0.1 mg using an electronic balance. In between mass measurements, samples should be kept in a lidded container to prevent contamination. If any measurement value is less than the detection limit of the balance, record the smaller of 0.1mg and the detection limit of the balance as the mass value.
- a. All individuals (and associated fragments) of a given taxon should be patted dry with a ChemWipe, placed a pre-combusted, weighed, and labeled weighboat. Measure the contents of the weighboat. Subtract the pan weight from this value to obtain the **Wet Mass** for that taxon.
 - b. Once wet mass has been measured, place the tissue+weighboat into a drying oven at 60°C for a minimum of 48hrs. After the minimum 48hrs, remove samples from the drying oven and allow to cool in a desiccator. Measure the contents of each weighboat. Subtract the pan weight from this value to obtain the **Dry Mass** for that taxon.
 - c. Once dry mass has been measured, place the tissue+weighboat into a muffle furnace at 550°C for 6hrs. Allow samples to cool in the oven and then move to a desiccator. Measure the contents of each weighboat. Subtract the pan weight from this value to obtain the ash mass. Subtract the ash mass from the dry mass to obtain the **Ash Free Dry Mass** for that taxon.

Inf fauna QA/QC

During sorting, a random, representative aliquot of every sample, representing at least 10% by volume of the sample, should be resorted by a different person (e.g., lab manager). The number of organisms found in the resorting should be recorded.

1. Sorting efficiency is calculated by:

$$\% \text{Efficiency} = 100 * \{ \# \text{orig} / [\# \text{orig} + (\# \text{resort} / \text{aliquot fraction})] \}$$

2. If sorting efficiency is less than 95%, then the entire sample should be resorted and checked again.
3. Organisms found in the re-sort should be given to the appropriate taxonomist for identification and enumeration for inclusion in the results from the sample.

During identification, samples should be organized in batches of 10 (or less). From each batch, 1 sample should be randomly selected and re-identified by a separate taxonomist, ideally from a separate laboratory. Initial-ID and re-ID data should be compared. The two taxonomists should come to a resolution on differences. After resolution, the number of individuals agreed as mis-identifications and the total number of individuals re-identified should be recorded. Differences in level of identification (e.g., species vs. genus) related to levels of experience between taxonomists or differences due damage of organisms from handling should not be considered errors.

1. Count accuracy is calculated as

$$\text{Count Accuracy} = \{1 - [\text{Absolute } (\# \text{ Individuals}_{\text{ReID}} - \# \text{ Individuals}_{\text{Original}}) / \# \text{ Individuals}_{\text{Resolved}}]\} * 100$$

2. Identification accuracy is calculated as

$$\text{Identification Accuracy} = [1 - (\# \text{ Individuals}_{\text{Mis-ID'd}} / \# \text{ Individuals}_{\text{ReID}})] * 100$$

3. If count accuracy or identification accuracy are less than 90% for a given batch, all samples within that batch should be re-identified and counted. Instances of lost individuals – where the re-identification counts fewer organisms than the initial identification and there are no obvious errors in differentiating anterior ends from posterior ends – should be noted and better attention to handling of specimens should be practiced by the initial taxonomist(s).

Step 23 – Sample storage and disposal

Step 24 – Data upload and entry

Appendices

Draft

1 - Infauna Sorting Sheet

Macrofauna Sorting Sheet

Station: _____	Taxonomy Laboratory: _____
Sorted by: _____	Sorting Laboratory: _____
Date Sorting Begins: / / mm/dd/yyyy Total time (hours): _____ # of Taxa Lots in Sample: _____ # of Sample Containers _____ Comments: _____ _____	
Quality Control Re-Sort Re-sorted by: _____ Date of re-sort: / / mm/dd/yyyy Percent Sorting Efficiency = $\{A / [A + (B/C)]\} * 100$ A = # of Organisms originally sorted: _____ B = # of Organisms found in resort: _____ C = Fraction of sample re-sorted (i.e., aliquot): _____ % Sorting Efficiency = _____	
Quality Control Actions: _____ _____ _____	
<i>Note: no action needed if sorting efficiency $\geq 95\%$</i> Signed: _____ Responsible Supervisor	

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